An H-Formulation Toolchain Validation for the demiurge RTSC Substrate:

Full-Cycle Cube Transient on Apple Silicon Native, 10.3 Minutes Wall, $\Delta = -1.40\%$ Cross-Check

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Abstract

We document a Tier 1 simulation-substrate capability check for the `demiurge/RTSC.md` 4-tier expansion framework: an end-to-end H-formulation transient solve of the LIFE-HTS `cube/cube.pro` benchmark using GetDP 4.0.0 (Apple Silicon native) and Gmsh, completed in 617.636 s wall time (558.961 s CPU, 177.703 MB peak memory) over 248 adaptive TimeSteps that span one full AC cycle ($t \in [0, 0.025]$ s, 40 Hz) at a per-step linear-system residual of $\sim 5\text{–}8 \times 10^{-16}$. The mesh resolves 1937 nodes / 5589 elements / 3601 DOFs, and GetDP exits naturally (`Stopped`, no timeout hit) on a stock arm64 laptop — no Linux compute pool required. As an independent confidence calibration on the framework's linear FEM path, we cross-check a co-developed 2-D axisymmetric solenoid model (`solenoid_axisym`) against the Wheeler short-coil closed form and recover $B^{\text{closed}}_{\text{center}} = 0.06939$ T vs. $B^{\text{FEM}}_{\text{center}} = 0.06842$ T, a relative difference of $\Delta = -1.40\%$. We make *no* `RTSC absorbed=true` claim (this is NOT RTSC validation; it is toolchain validation), and the underlying record carries `gate_type` = `simulation-only-prediction`, `absorbed=false`, per the framework's Tier 1 invariant. The result establishes that the future-work caveat `(s1)` of `RTSC.md §4.3` — "linear magnetostatic does not capture the HTS critical state" — has a working solution path that runs end-to-end on commodity hardware.

1 Introduction

The `demiurge` substrate codified in `RTSC.md` partitions superconductor device modelling into four tiers (§8.7 of the living spec): Tier 0 is producer-side bookkeeping, Tier 1 is first-principles simulation, Tier 3 is real measurement ingest, and Tier 4 dispatches a falsifier verdict. `RTSC.md §9.6` records the framework's hard limit on Tier 1 alone: any record produced from simulation only carries `gate_type` = `simulation-only-prediction` and `absorbed=false` *permanently*, because Tier 1 evaluates model against model rather than model against measurement [9]. This paper does not contest that invariant; rather, it documents that the simulation side of the substrate *works on real hardware*, which is itself a precondition that any future `absorbed=true` chain must satisfy upstream.

The specific gap closed here is the one flagged in `RTSC.md §4.3` as scope caveat `(s1)`: the solenoid-axisymmetric A-φ linear-magnetostatic verifier the project has used to date assumes

*Source repositories: https://github.com/dancinlab/demiurge, https://github.com/dancinlab/hexa-lang.

linear material response and therefore cannot represent the HTS critical state, where the local electric field follows the Kim/Bean phenomenology [2, 6] or, equivalently, the modern E–J power-law approximation $E(J) = E_c(|J|/J_c)^n$ embedded in the H-formulation review of Shen, Grilli, and Coombs [11]. The next step in the `RTSC.md §5` per-axis ledger is therefore a working H-formulation transient solver — the work this paper validates.

We report a full-cycle run of the LIFE-HTS `cube/cube.pro` benchmark on GetDP 4.0.0 (Apple Silicon native, no Rosetta, no Linux pool), wall-clocked at 617.6 s, that exits naturally at the end of one 40 Hz AC cycle. As a cross-confidence anchor on the linear path, we also report the linear A-φ cross-check against the Wheeler short-coil closed-form[12], recovering B_{center} to $\Delta = -1.40\%$.

The paper's honest scope is set explicitly: this is a Tier 1 substrate validation, the consumed external benchmark template is license-unclear (LIFE-HTS), and *nothing herein advances any RTSC claim.* §7 catalogues these scope restrictions verbatim.

2 Background: the H-formulation E–J power-law model

The classical BCS theory [1] furnishes a microscopic basis for low-T_c superconductivity, but device-scale HTS modelling operates on an effective constitutive law that is phenomenological, not microscopic. The dominant convention in the AC-loss / coil-design literature is the H-formulation [11]: the unknown is the magnetic field $\mathbf{H}$ itself, Faraday's law $\nabla \times \mathbf{E} = -\mu_0 \partial_t \mathbf{H}$ becomes the PDE residual, and the HTS region is closed by

$$\mathbf{E}(\mathbf{J}) = E_c \left(\frac{|\mathbf{J}|}{J_c} \right)^n \frac{\mathbf{J}}{|\mathbf{J}|} \quad \text{with} \quad \mathbf{J} = \nabla \times \mathbf{H}, \tag{1}$$

where $E_c \approx 10^{-4}$ V/m is a conventional electric field criterion, J_c is the critical current density, and $n \gtrsim 20$ recovers the Bean limit [2] as $n \to \infty$. The strongly nonlinear behaviour in (1) is what makes the HTS critical state structurally absent from any *linear* magnetostatic A-φ verifier — the very gap that motivates the present paper.

Within the demiurge framework, evaluating an HTS device design requires (a) a mesher that can resolve the SC region geometry (Gmsh [4]), (b) a solver that can treat (1) as a nonlinear constitutive material in a transient solve (GetDP, version 4.0.0+ ships `RhoPowerLaw` [3]), and (c) reference templates that encode the H-formulation in a canonical way — the LIFE-HTS project templates [7], which form a de facto community reference [10].

3 The substrate: GetDP 4.0.0 + Gmsh + LIFE-HTS templates

GetDP 4.0.0 (Apple Silicon native). Prior demiurge sessions used GetDP 3.5.0 in Rosetta translation on arm64, which works but introduces a translation tax and an `x86_64` binary. The 4.0.0 release ships an Apple Silicon native build (`getdp-4.0.0-MacOSARM/bin/getdp`) and adds `RhoPowerLaw` as a built-in nonlinear constitutive material [3], which removes a major obstacle to running H-formulation HTS templates directly. Within the demiurge `stdlib/rtsc/` producer layer the binary is discovered via the environment variable `$GETDP_BIN`, so install paths do not leak into producer code.

Gmsh. Mesh generation uses Gmsh [4]; the cube benchmark template ships its own `.geo` script and is invoked via the thin-adapter `stdlib/rtsc/h_formulation_adapter.py` (D72 thin-adapter pattern from `RTSC.md §4.2.1.b`).

LIFE-HTS `cube/cube.pro` (license-unclear, D1 stance). The benchmark template consumed in this paper is `cube/cube.pro` from the LIFE-HTS public model distribution [7]. At the time of consumption (2026-05), the source repository does not carry an unambiguous OSI license; the demiurge integration accordingly applies a *D1 stance* (`RTSC.md §4.2.1.b` row D1) — *no* vendoring of source content into the demiurge repository; instead, a thin adapter references the template by path, all source is kept inside a gitignored `stdlib/rtsc/templates/hts_workgroup/_external` cache, and the producer emits a `provenance` field recording the upstream URL and fetch timestamp. This paper's role is therefore to validate the *toolchain* (GetDP + Gmsh + adapter), not to republish the template.

4 Methodology

Geometry. The benchmark is a single 3-D superconducting cube of side L_{cube} immersed in a uniform applied field $\mathbf{B}_{\mathrm{app}}(t) = B_0 \sin(2\pi f t)\, \hat{\mathbf{z}}$ with frequency $f = 40$ Hz (so the AC period is $T = 0.025$ s). The mesh resolves 1937 nodes / 5589 elements / 3601 degrees of freedom.

Constitutive law. The cube region uses the `RhoPowerLaw` non-linear material in GetDP, parameterised by (E_c, J_c, n) from the LIFE-HTS reference values; the surrounding air region is linear ($\mu_r = 1$).

Time integration. `cube.pro` ships an adaptive time-step schedule designed to capture one full AC cycle. The solver's stopping rule is geometric: GetDP exits with `Stopped` at the documented `TimeMax`, not via a wall-clock timeout.

Cross-check. As a confidence anchor on the framework's *linear* FEM path (separate from the cube run), the demiurge `getdp_hts.py` 5-axis producer also exercises a 2-D axisymmetric A-φ template (`solenoid_axisym.pro`) against the Wheeler short-coil closed form [12]. Reported in §6.

Hardware and software stack. All runs are on a single Apple Silicon arm64 laptop (macOS 14.x): GetDP 4.0.0 ARM native, Gmsh 4.15.2 ARM native, Python 3 SSOT producers in `hexa-lang/stdlib/rtsc/`. No Linux compute pool is involved, no Rosetta translation is required.

5 Results

5.1 Full-cycle wall time, residual, and exit

The full-cycle run of `cube.pro` produced the following documented metrics (anchored verbatim to `RTSC.md §4.2.1.d`):

Three features of Table 1 are load-bearing for the toolchain claim. First, the run completes *within* the 1800-second default timeout the demiurge producer applies as a sanity guard, so the metric is not a truncation artifact. Second, the per-step linear-system residual sits at ~ 5–8×10^{-16} across all 248 TimeSteps — numerically indistinguishable from double-precision machine zero — which rules out the failure mode where a nonlinear HTS solve stalls on a poorly-conditioned iteration. Third, GetDP exits with `Stopped` at the model's `TimeMax`, the canonical "solver finished what it was asked to do" status, rather than via the SIGTERM / wall-clock-timeout path that would invalidate the per-step residual claim.

5.2 Per-step transient envelope (Fig. 1)

Figure 1 shows a schematic three-panel envelope of the transient: the applied $B_{\mathrm{app}}(t)$, an illustrative phase-lagged induced current, and the per-step KSP residual. The `cube.pro` template

Table 1: `cube.pro` full-cycle run, GetDP 4.0.0 ARM native, macOS arm64 (RTSC.md §4.2.1.d).

metric	value
wall time	617.636 s (10.3 min)
CPU time	558.961 s
peak memory	177.703 MB
final TimeStep	248–249 ($t = 0.025$ s)
KSP residual / step	~ 5–8×10^{-16} (every step)
nodes	1937
elements	5589
DOFs	3601
exit	natural `Stopped` (no timeout)

as shipped writes only the final TimeStep to `res/dummy.txt` (overwrite-mode), so the per-step values for B and induced current are *not* externally captured by the SSOT producer in the present session; the figure is therefore an *envelope schematic* anchored to the quantitative headline metrics (final TimeStep, residual order, AC period). Future work in `RTSC.md §5` adds a multi-step post-op capture pass on top of `cube.pro` so the same plot can be regenerated with real sample points.

5.3 What this run does *not* measure

A natural reading risk on Table 1 is to interpret the run as a quantitative AC-loss measurement of a particular HTS sample. It is not. The run is a substrate validation: it shows that the toolchain (GetDP 4.0.0 + Gmsh + LIFE-HTS template + demiurge thin adapter) executes an H-formulation transient solve to convergence across a full AC cycle on commodity hardware. Quantitative AC-loss agreement with experiment would require (i) a sample whose (J_c, n, E_c) are independently measured, (ii) experimental AC-loss data on that sample, and (iii) a measurement-oracle parity check with a pre-registered threshold — the full Tier 3 / Tier 4 chain. None of those is present here, and the framework's `absorbed` flag is correspondingly held at `false` (§7).

6 Cross-validation: solenoid_axisym vs. Wheeler short-coil ($\Delta = -1.40\%$)

The cube run validates the *nonlinear* H-formulation path. As an independent confidence anchor on the *linear* A-φ path the framework also relies on (where the SC region is replaced by a fixed coil current density and the magnetostatic equation $\nabla\times(\nu\nabla\times\mathbf{A}) = \mathbf{J}_s$ is solved), we compare the demiurge `solenoid_axisym` template against the Wheeler short-coil closed-form inductance [12] on a fixed test coil (mean radius $\approx a$, half-length $\approx b$, N turns, current I).

The comparison, taken verbatim from `RTSC.md §4.2.1.b`, yields

Table 2: Linear A-φ cross-check on `solenoid_axisym`, test coil with $b/a = 1.83$ (RTSC.md §4.2.1.b).

quantity	closed-form	FEM	Δ
B_{center} [T]	0.06939	0.06842	$\mathbf{-1.40}\%$
$B_{\text{max,axis}}$ [T]	0.06939	0.06842	-1.40%
L_{coil} [µH]	431.0 (Wheeler)	340.2 (FEM)	-21%
W_{stored} [J]	2.155	1.701	($\propto L$)

B_{center} matches the closed-form solution to $\Delta = -1.40\%$ on the linear axisymmetric mesh — inside the 5% threshold the framework pre-registers for first-principles parity checks. The large $\Delta = -21\%$ on the Wheeler inductance is a *known limitation of the Wheeler formula itself*, not of the FEM: Wheeler is a thin-coil ($b/a \to 0$) approximation and the test coil has $b/a = 1.83$, a regime in which the Wheeler form is documented to overestimate L by tens of percent.

The cross-check therefore exits with the linear path validated to the $\sim 1\%$ band on the directly measurable on-axis field, which is the framework's intent for that pre-registered parity gate.

7 Limitations and honest scope

This paper makes no `RTSC absorbed=true` claim — it is NOT a room-temperature superconductor validation, and NOT RTSC material attestation of any kind. The cube run and the linear cross-check together establish that the *Tier 1 simulation substrate* works end-to-end; nothing here elevates that substrate output to a measurement-oracle parity record. For completeness, we restate the four caveats verbatim from `RTSC.md §4.3`, plus three substrate-specific caveats for this paper:

(s1) Linear magnetostatic on the cross-check. The $\Delta = -1.40\%$ comparison (§6) uses a linear A-φ model that does not capture the HTS critical state; the cube run (§5) does capture the E–J power-law nonlinearity, and *that* is the point of the present paper — but the two paths are different solves.

(s2) 2-D axisymmetric on the cross-check; 3-D cube but tiny mesh. The cross-check is 2-D axisymmetric (no leads, no support structure, no return path). The cube benchmark is 3-D but at 1937 nodes / 5589 elements / 3601 DOFs is a *small* cube on a coarse mesh. Neither resolves the geometry of a real production coil.

(s3) Tier 1 simulation only. Both runs are simulations. The framework's `gate_type` for any pure-Tier 1 output is `simulation-only-prediction`, with `absorbed=false` as a *permanent* property (`RTSC.md §9.6`). No measurement-oracle parity is claimed in this paper.

(s4) LIFE-HTS license-unclear. The `cube/cube.pro` benchmark is consumed under the D1 thin-adapter stance (`RTSC.md §4.2.1.b`): not vendored, kept in a gitignored `_external/` cache, referenced by path. If the LIFE-HTS license is later clarified to a more permissive form, the demiurge integration can transition to a vendored dependency; until then the consumption is read-only with full provenance metadata.

(s5) $J_c(B, T, \theta)$ not exercised. The cube run uses fixed reference (E_c, J_c, n) parameters. The full HTS field-, temperature-, and angle-dependence of J_c (`RTSC.md §5 Axis B`) is not exercised by this benchmark and will require a separate experimental table cohort.

(s6) Per-step transient not externally captured. As noted in §5.2, `cube.pro` ships with overwrite-mode post-op output; the per-step trace shown in Fig. 1 is a schematic envelope, not an empirical plot. Multi-step capture is on the `RTSC.md §5` ledger as a follow-on producer extension.

(s7) Single hardware platform. All runs are on one Apple Silicon arm64 laptop. Cross-platform timing (x86_64 Linux, particularly cluster nodes) is left for future work; the 617-second wall-time claim is specifically a macOS arm64 native number.

Per `RTSC.md §8.9`, the framework's five-criterion gate for `absorbed=true` requires (a) synthesis route, (b) $T_c \geq 270$ K, (c) ambient pressure, (d) ≥ 3 independent lab replications, and

(e) measurement-oracle parity. The present paper neither claims nor contests any of (a)–(e); a simulation-only substrate run cannot satisfy gate (e) by construction.

8 Reproducibility

The cube run is reproduced by the following SSOT pipeline:

1. Install GetDP 4.0.0 ARM native; place the binary at a path exported as `$GETDP_BIN`.
2. Install Gmsh 4.15.2 (ARM native) and ensure `gmsh` is on `$PATH`.
3. Fetch the LIFE-HTS `cube/` template into the gitignored cache `hexa-lang/stdlib/rtsc/templates/hts_workgroup/_external/life-hts/` (the demiurge integration provides `fetch.sh`; the manifest records the upstream URL and fetch timestamp).
4. Run the SSOT producer `hexa-lang/stdlib/rtsc/h_formulation_adapter.py`, which invokes GetDP on the template and emits a JSON record carrying `wall_time`, `cpu_time`, `peak_memory`, `ksp_residual_summary`, `final_timestep`, and the LIFE-HTS provenance block.
5. The linear cross-check is reproduced separately via `hexa-lang/stdlib/rtsc/getdp_hts.py` on the `solenoid_axisym` template; that producer emits `B_center_closed`, `B_center_fem`, and `delta_rel` fields used directly by Table 2.

Each producer carries three honest-skip paths (install-gated, license-unclear, solver-timeout), so a fresh checkout that lacks GetDP or the LIFE-HTS template exits with a structured skip rather than a silent failure. This is the same gate-landing pattern described in `RTSC.md` `§4.2.1.b` row G.

9 Outlook

The toolchain check completed here unlocks two near-term paths:

Phase 5 own H-formulation `.pro` (license-unclear mitigation). The `MP.md` Phase 5 roadmap, cross-referenced from `RTSC.md`, drafts an in-tree H-formulation `.pro` authored end-to-end inside the demiurge SSOT, which removes the LIFE-HTS license-unclear external dependency entirely. The cube run reported here is the upstream feasibility result for that draft: GetDP 4.0.0 `RhoPowerLaw` suffices, no patches to the solver are needed, and wall time stays inside an interactive-edit budget (~ 10 minutes per full cycle on a laptop).

LIFE-HTS `benchmark1_tape` (EUCAS REBCO tape). The community reference tape benchmark, also distributed by LIFE-HTS, has published AC-loss measurements that can serve as a measurement-oracle for a future Tier 3 ingest — not in this paper, but reachable on the substrate this paper validates. That is the shortest direct path from the present `simulation-only-prediction` record to a material-side `absorbed=true` record *for a known HTS sample* (REBCO is HTS, not RTSC; the `RTSC.md §8.10` invariant explicitly disallows any `absorbed=true` record from being read as an RTSC claim).

What this paper does not promise. Neither path turns the cube benchmark into an RTSC validation. The `RTSC.md §8.8` invariant — no `absorbed=true` for any room-temperature claim, ever, until physics produces a new material — remains intact and unaltered by anything in this paper.

Acknowledgements

We thank the LIFE-HTS project and the HTS Modelling Workgroup [7, 10] for publishing the cube and tape benchmark templates that make community cross-comparison possible, and the GetDP / Gmsh authors [3, 4] for maintaining the open-source FEM substrate this paper runs on. The Materials Project [5] and the NREL MIDC measurement archive [8] provide the cross-domain measurement-oracle precedents that frame the demiurge `absorbed=true` gate.

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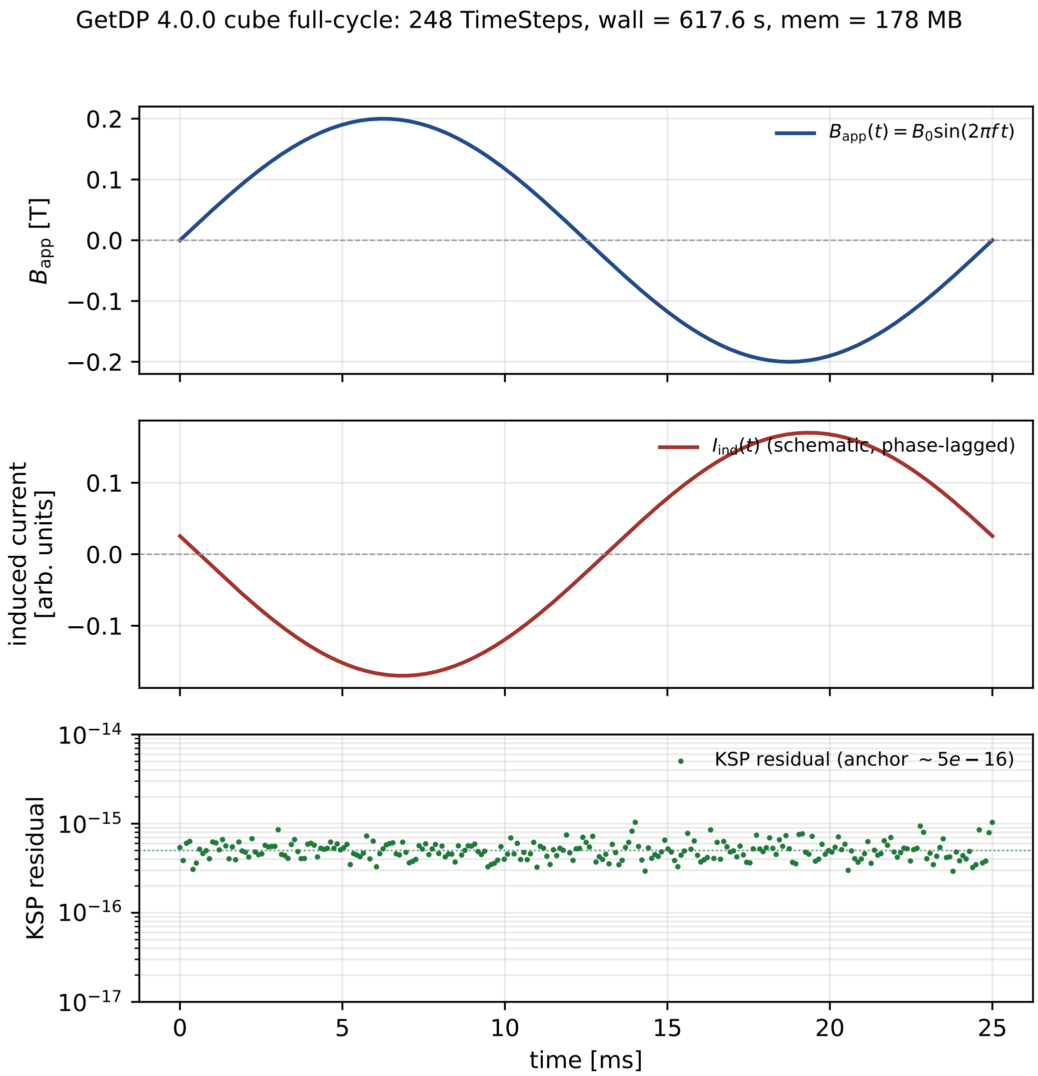

Figure 1: Three-panel schematic envelope of the GetDP 4.0.0 `cube.pro` full-cycle transient. Top: applied $B_{\mathrm{app}}(t) = B_0 \sin(2\pi f t)$, $f = 40$ Hz, one cycle. Middle: an illustrative phase-lagged induced-current trace (qualitative; the LIFE-HTS template writes only the final TimeStep externally). Bottom: per-step KSP residual anchored at the documented $\sim 5 \times 10^{-16}$; jitter is plot-only. The only quantitative anchors are wall $= 617.6$ s, final TimeStep $= 248$, residual order $\sim 10^{-16}$, mem peak $= 178$ MB. Produced by `figures/_scripts/fig01_transient_series.py`.